## §1 Revised inquiry and response to feedback

### Inquiry question

In a simulated platform based on YouTube, to what extent can visibility boosts granted after reinstatement compensate for the exposure opportunities lost during a video's removal?

### Current boundary

This project examines how much YouTube videos lose presentation opportunities while removed and whether a temporary visibility boost after reinstatement can compensate for those missed opportunities.

The real-world arrangement involves creators, moderators, and viewers, as well as YouTube's recommendation algorithm and Community Guidelines appeals process. The model, however, abstracts many of these actors and focuses on removal, reinstatement, and visibility of a video.

### Revision from Deliverable 1

There were three significant revisions:

1. The platform reference is now explicit. The project is grounded in YouTube's Community Guidelines removal-and-appeal process.
2. The relationship between availability, governance and visibility is clearer. These are represented in the model by video availability status, moderator and reinstatement rules, and the presentation-selection-rule, respectively.
3. A new evidence source was added: an academic article discussing content moderation on YouTube

### Reason for the revision

The revisions are mainly a response to the feedback requesting a clearer platform referent and a more precise boundary. In addition, both YouTube's official documentation and external research helped clarify which aspects of the real-world platform should be represented in the model and which could be simplified or excluded.

## §2 Model plan, evidence, and targeted DEL02 slice

### §2.1 Broader model plan

The broader model represents video removal, successful appeal and reinstatement, and the effect these changes in availability have on the presentation opportunities videos receive.

## What is represented

The model is informed by YouTube's documented processes but uses video agents and simplified rules to represent selected relationships. It does not duplicate YouTube's actual systems.

| YouTube arrangement | Model representation | Role in the model |
|---|---|---|
| Creators | Synthetic video agents | Represent uploaded videos and store attributes such as content category, availability, selection weight, and presentation count. |
| Moderators | Moderator rule | Determines which videos are removed and for how long, considering their assigned content categories. |
| Appeals Process | Reinstatement rule | Assumes every removed video is successfully appealed and makes it available again after its assigned removal duration. |
| Recommendation algorithm | Presentation-selection-rule | Selects available videos for presentation slots using their selection weights. |
| Viewers | Presentation slots | Individual viewers are not represented in the model. Assume a single non-logged in viewer with 100 presentation slots daily. A video is chosen for each of the slots by the presentation-selection-rule. |

## Assumptions

In the simulation, the following are assumed for simplicity. They don’t represent how YouTube’s content moderation and recommendation algorithm function:

- All removals are successfully appealed
- Removal duration is predefined
- All available videos compete for the same presentation slots

## What is not represented

| YouTube arrangement or feature | Reason it is not represented |
|---|---|
| Observers reporting videos | Omitted.<br><br>Reporting behaviour is not simulated. The moderator rule removes videos based on the video categories. |
| Views, Impressions, and Audience Engagement | Omitted.<br><br>The inquiry focuses on whether a video is presented, not whether viewers watch, click, or interact with it. |

| | |
|---|---|
| Watch history, search history, channel subscriptions, and other recommendation signals | Simplified.<br><br>The model uses synthetic content-categories and calculated weights to represent differences in selection likelihood. These replace the more complex signals used by YouTube and do not claim to reproduce its recommendation process. |

## Parameter and comparison conditions

The single parameter changed between the baseline and alternative is the boost-multiplier (which affects the selection weight).

| Condition | Removal | After reinstatement |
|---|---|---|
| No-removal reference | No videos are removed | N/A |
| Standard reinstatement (baseline) | Some videos are removed on a schedule | *1x multiplier*<br>i.e. weight returns to what it was before |
| Temporary restoration (alternative) | Same videos are removed on the same schedule | *2x multiplier*<br>i.e. weight temporarily increases before returning to what it was before |

## Intended output

The model will record:

- The number of presentation slots received by the reinstated videos under each condition
- Whether the boost increases the number of slots received after reinstatement (baseline vs alternative)
- Whether the boost helps a reinstated video recover opportunities it missed (reference vs alternative)

## Evidence and Source table

| Source | Role | Access and use | Status |
|---|---|---|---|
| YouTube Help, Appeal a Community Guidelines strike or video removal. Official Page | Official description of YouTube's removal, appeal, and reinstatement process. Used to understand the phenomenon and inform the simulation's simplified version | Publicly available | Observed |
| YouTube Help, How YouTube recommendations work. Official Page | Establishes that YouTube recommendations are personalized and consider multiple signals. The | Publicly available | Observed |

| | | | |
|---|---|---|---|
| | model's presentation-selection-rule is a simplification. | | |
| Research Article, Aspirational platform governance. Article doi | Discusses creators' perception that certain types of content are treated unfairly through content moderation. Informs the decision to include content categories in the moderator rule. | Publicly available | Inferred.<br><br>Category-based removal is just an observation of creators, not a confirmation of actual YouTube enforcement patterns |

## §2.2 Targeted DEL02 slice

Instead of implementing the full model, a smaller version was developed for this deliverable.

### What's included: DEL02 slice vs Deferred Version

This table outlines how the model is designed in the DEL02 slice and what is deferred to the later version.

| Model element | DEL02 slice version | Deferred version |
|---|---|---|
| Synthetic video agents | A Mesa agent that represents an uploaded video and stores all video attributes | already fully developed in DEL02 version |
| Moderator rule | The removed videos are pre-chosen | Moderator agent that removes videos based on content categories |
| Reinstatement rule | Gives a boost of x2 on Days 4 and 5 | Flexible reinstatement rule that determines boost value and duration based on removal duration |
| Presentation-selection-rule | A function that decides which videos get chosen for each presentation slot | already fully developed in DEL02 version |
| Presentation slots | To represent visibility | already fully developed in DEL02 version |

The simulation was also run using smaller numbers:

| Simulation element | DEL02 slice version | Deferred plan |
|---|---|---|
| Video agents | 10 | 50 |
| Simulation days | 5 | 7 |
| Presentation slots per day | 20 | 100 |
| Removed videos | V01, V02, V06 | Determined by moderator agent |

| | | |
|---|---|---|
| Content Categories | 2 (CategoryA and CategoryB) | Multiple content categories, videos assigned >1 category |
| Outputs | CSV and graphs | Browser visualization |

### Parameter and Comparison Condition

In the DEL02 slice, the single parameter changed between the baseline and alternative is the boost-multiplier (which affects the selection weight).

| Condition | What happens |
|---|---|
| No-removal reference | No videos are removed |
| Standard reinstatement (baseline) | The selected videos are removed on Day 2 and reinstated on Day 4. Their selection weights return to its original values |
| Temporary restoration (alternative) | The same videos are removed and reinstated but receive 2x their original weights after reinstatement. |

### Why this is an appropriate first step

The DEL02 slice uses fewer video agents, a shorter simulation period, and a simpler moderator rule than the broader model. It provides a runnable version of the main components and so makes the effects of removal, reinstatement, and temporary restoration observable while also making clear what could be changed and expanded on in the later version.

### DEL02 slice Output

The DEL02 slice produces raw data for each condition, recording the replication number, condition (reference, baseline, or alternative), simulation day, and total presentation slots received by the three affected videos.

These results are then analysed to calculate averages across replications and compare the conditions. See Section 4 for details of the outputs.

## §3 Mesa model implementation summary

**Source example.** Mesa's Epstein Civil Violence example was the starting point for the model. It provided a code structure for creating agent types, designing model conditions and advancing through the simulation.

**Current run**. The model counts how times a video is selected for presentation. It compares the count for a video removed then reinstated (baseline) with the count of a video removed then reinstated with a visibility boost (alternative).

**Codebase reference.**

Repository URL: https://github.com/funmbia/visibility-restoration-modeling.git

DEL02 Release URL: https://github.com/funmbia/visibility-restoration-modeling/releases/tag/DEL02

## §4 Repository outputs: descriptions and comparison

This section discusses the outputs produced by DEL02 slice. The model has 3 conditions (reference, baseline, and alternative). The value being measured throughout is the number of presentation slots the 3 affected videos receive.

*Note: Files comparing all three conditions use the* ***`ALL`*** *prefix.*

| Output files | Description/ What it contains |
|---|---|
| `reference_raw_00_daily_presentation_slots.csv` | Daily presentation slots received by the 3 affected videos when they remain available throughout the simulation |
| `baseline_raw_00_daily_presentation_slots.csv` | Daily presentation slots received by the affected videos when reinstated without a boost |
| `alternative_raw_00_daily_presentation_slots.csv` | Daily presentation slots received by the affected videos when reinstated with a 2x boost multiplier |
| `ALL_processed_00_daily_presentation_slots.png` | Line graph showing the daily averages across all three conditions |
| `ALL_processed_01_combined_presentation_slots.png` | Bar graph showing the five-day averages across all three conditions |
| `ALL_processed_02_summary_list.csv` | Summary of the five-day averages as well as calculations of the differences between the three conditions |

### Conditions that produced the outputs

The DEL02 slice uses 10 video agents over 5 simulated days with 20 presentation slots each day. Five videos belong to Category A with a selection weight of 1.0, and five belong to Category B with a selection weight of 2.0. Three videos (V01, V02, and V06) are chosen as the affected videos.

The model compares three conditions:

1. **Reference:** No videos are removed throughout all 5 days. This is necessary to see how many presentation opportunities the videos would have gotten.
2. **Baseline:** The three affected videos are removed on Day 2 (their weight set to 0) and reinstated on Day 4 (their weight set back to the original values)
3. **Alternative**: The 3 affected videos are removed on Day 2 (their weight set to 0) and reinstated on Day 4 (their weight set to 2x the original values)

## What is visible in the outputs

The first three outputs in the table above contain daily results across all 10 replications. The '*total_presentation_slots'* field records the combined number of presentation slots received by the three videos each day.

- In **`reference_raw_00_daily_presentation_slots.csv`**, the affected videos remain available throughout the simulation and continue receiving presentation slots

- In both **`baseline_raw_00_daily_presentation_slots.`**csv and **`alternative_raw_00_daily_presentation_slots.csv`**, they receive zero slots on Days 2 and 3 because they are unavailable.

## Comparison

Comparing the last columns of b**`aseline_raw_00_daily_presentation_slots.csv`** and **`alternative_raw_00_daily_presentation_slots.csv`**, we see that on Days 4 and 5 (after reinstatement), the affected videos generally receive more presentation slots in the alternative condition than the baseline.

The daily values are averaged and graphed in **`ALL_processed_00_daily_presentation_slots.png`** to make this pattern easier to compare. In this graph, it's clear that the alternative has higher values than the baseline on Days 4 and 5.

**`ALL_processed_01_combined_presentation_slots.png`** graphs the averages for all 5 days per condition. This file makes the overall differences across the conditions clear.

## Connection to the Inquiry

The outputs show that the temporary boost increases the presentation opportunities received by the affected videos after reinstatement. However, as seen in **`ALL_processed_01_combined_presentation_slots.png`**, it does not fully make up for the opportunities missed during removal.

**`ALL_processed_02_summary_list.csv`** reports a recovery percentage of 49.52%**.** This means the alternative provides halfway rather than full recovery of missed presentation opportunities.

## §5 Interpretation, limitations, and next steps

### What the model shows.

Under the conditions and rules of the DEL02 model, applying a 2x selection-weight boost to reinstated videos increases their presentation slots compared with the baseline. However, the boost does not fully compensate for the removal because it recovers only part of the opportunities missed during the removal.

**What this helps investigate.** The model helps investigate whether increasing a video's selection weight after reinstatement can make up for the presentation opportunities it missed while removed. By changing the boost size or duration, I can test how much recovery it produces.

**What this model doesn't establish.** These results do not show how YouTube recommends videos or how real users respond to reinstated content. They also do not establish the real-world effect of removal, whether all successfully appealed videos experience the same outcome, or whether a 2x boost would have the same effect on YouTube.

### Limitations and Uncertainty

In addition to the assumptions listed in **Section 2.1**, the presentation-selection rule and presentation slots are simplified representations of visibility and recommendation. Selection is also performed with replacement, meaning the same video can receive multiple presentation slots in one day.

### Next Steps

- Implement the outlined elements deferred, as listed in **Section 2.2**
- Test longer boost durations to see whether it improves recovery
- Test multipliers higher than 2x to see how the recovery changes